Untitled

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PART A: Basic Operations

A1

```
v <- seq(5, 50, 5)

print(v)
```

```
##  [1]  5 10 15 20 25 30 35 40 45 50
```

```
print(length(v))
```

```
## [1] 10
```

```
print(sum(v))
```

```
## [1] 275
```

```
print(mean(v))
```

```
## [1] 27.5
```

A2

```
w <- rep(c(1, 2, 3), times = 4)
z <- rep(c(1, 2, 3), each = 4)

print(w)
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
print(z)
```

```
##  [1] 1 1 1 1 2 2 2 2 3 3 3 3
```

A3

```
remainder <- 17%%5
remainder
```

```
## [1] 2
```

```
division <- 17/2
division
```

```
## [1] 8.5
```

```
exponent <- 2^10
exponent
```

```
## [1] 1024
```

```r
print(sqrt(144))
```

```
## [1] 12
```

```r
print(round(pi, 3))
```

```
## [1] 3.142
```

A4

```r
a <- c(2, 4, 6, 8)
b <- c(1, 3, 5, 7)

a+b
```

```
## [1]  3  7 11 15
```

```r
a*b
```

```
## [1]  2 12 30 56
```

```r
a>b
```

```
## [1] TRUE TRUE TRUE TRUE
```

```r
sum(a*b)
```

```
## [1] 100
```

A5

```r
print(paste("Student-1", "Student-2", "Student-3"))
```

```
## [1] "Student-1 Student-2 Student-3"
```

```r
print(typeof(42))
```

```
## [1] "double"
```

```r
print(typeof(42L))
```

```
## [1] "integer"
```

```r
print(typeof("42"))
```

```
## [1] "character"
```

```r
print(typeof(41i))
```

```
## [1] "complex"
```

```r
print(typeof(TRUE))
```

```
## [1] "logical"
```

PART B: Data Fraames and Conditional Statements

B1

```r
students <- data.frame(
  id = 1:8,
  name = c("Alice", "Bruno", " Chen", "Dana", "Elif", "Farid", "Grace","Hiro"),
  sex = c("F", "M", "M", "F", "F", "M", "F", "M"),
  section = c("A", "B", "C", "B", "B", "C", "A", "B"),
  midterm = c(78, 85, 92, 67, 74, 88, 95, 59),
```

```
  final = c(82, 79, 96, 71, 80, 91, 93, 65),
  attendance = c(95, 88, 100, 72, 90, 85, 98, 60),
  stringsAsFactors = FALSE
)
```

B2

```
str(students)
```

```
## 'data.frame':    8 obs. of  7 variables:
##  $ id        : int  1 2 3 4 5 6 7 8
##  $ name      : chr  "Alice" "Bruno" " Chen" "Dana" ...
##  $ sex       : chr  "F" "M" "M" "F" ...
##  $ section   : chr  "A" "B" "C" "B" ...
##  $ midterm   : num  78 85 92 67 74 88 95 59
##  $ final     : num  82 79 96 71 80 91 93 65
##  $ attendance: num  95 88 100 72 90 85 98 60
```

```
dim(students)
```

```
## [1] 8 7
```

```
names(students)
```

```
## [1] "id"         "name"       "sex"        "section"    "midterm"
## [6] "final"      "attendance"
```

```
summary(students)
```

```
##        id              name               sex             section     
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :59.00
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:72.25
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :81.50
##  Mean   :4.50   Min.nchar:4   Min.nchar:1   Min.nchar:1   Mean   :79.75
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00
##  Max.   :8.00                                           Max.   :95.00
##      final         attendance
##  Min.   :65.00   Min.   : 60.00
##  1st Qu.:77.00   1st Qu.: 81.75
##  Median :81.00   Median : 89.00
##  Mean   :82.12   Mean   : 86.00
##  3rd Qu.:91.50   3rd Qu.: 95.75
##  Max.   :96.00   Max.   :100.00
```

B3

```
students$average <- (students$midterm + students$final) /2

 print(students)
```

```
##   id  name sex section midterm final attendance average
## 1  1 Alice   F       A      78    82         95    80.0
## 2  2 Bruno   M       B      85    79         88    82.0
## 3  3  Chen   M       C      92    96        100    94.0
## 4  4  Dana   F       B      67    71         72    69.0
## 5  5  Elif   F       B      74    80         90    77.0
## 6  6 Farid   M       C      88    91         85    89.5
## 7  7 Grace   F       A      95    93         98    94.0
```

```
## 8  8  Hiro   M       B      59    65         60    62.0
```